



Guadalupe Valencia Toxqui, Ph.D.

Bacteriophage Biologist · Center for Phage Technology, Texas A&M University

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Research Position

2023–Present **Postdoctoral Researcher**, Ramsey Lab, Center for Phage Technology, Department of Biology, Texas A&M University. *Phage Mu lysis timing, biofilm-disrupting phages, and phage genome engineering.*

Education

2016–2022 **Doctorate of Science**, Genetics & Molecular Biology, CINVESTAV, Mexico City. *Thesis: Early transcription antitermination of phage mEp021 regulated by the Gp17 (N-like) protein. Supervisor: Luis Kameyama.*

2014–2016 **Master of Science**, Genetics & Molecular Biology, CINVESTAV, Mexico City. *Thesis: ORFan gp26 characterization of E. coli bacteriophage mEp021. Supervisor: Luis Kameyama.*

2008–2013 **Bachelor of Biology**, Meritorious Autonomous University of Puebla (BUAP). *Thesis: Monitoring & molecular detection of enteropathogenic E. coli from sewage water capable of degrading an azo dye. Supervisor: Edith Chavez Bravo.*

Peer-Reviewed Publications (Valencia-Toxqui in bold; * equal contribution)

Valencia-Toxqui G, Sugumar S, Ramsey J. (2025) Isolation and characterization of biofilm-disrupting Proteus phage Premi. *Scientific Reports* 15, 39780. doi:10.1038/s41598-025-23545-3

Ripberger P, **Valencia-Toxqui G**, Ramsey J. (2025) The complete genome of Escherichia phage Midge. *Microbiology Resource Announcements* 14:e00861-25. doi:10.1128/mra.00861-25

Negrete-Méndez H*, **Valencia-Toxqui G***, Sepúlveda-Robles OA, Ríos-Castro E, Hurtado-Cortés JC, Flores V, Cázares A, Kameyama L, Martínez-Peñañiel E, Fernández-Ramírez F. (2025) Genomic and proteomic analyses of Nus-dependent non-lambdoid phages reveal a novel coliphage group prevalent in gut: mEpimml. *Frontiers in Microbiology* 16:1480411. doi:10.3389/fmicb.2025.1480411

Negrete-Méndez H, **Valencia-Toxqui G**, Martínez-Peñañiel E, Medina-Contreras O, Fernández-Ramírez F, Morales-Ríos E, Navarro-González LJ, Torres-Flores JM, Kameyama L. (2025) A Lambda-evo (λevo) phage platform for Zika virus EDIII protein display. *Applied Microbiology and Biotechnology* 109(1):8. doi:10.1007/s00253-024-13380-3

Valencia-Toxqui G, Ramsey J. (2024) How to introduce a new bacteriophage on the block: a short guide to phage classification. *Journal of Virology* 98:e01821-23. doi:10.1128/jvi.01821-23

Valencia-Toxqui G, Ballinas-Turrén EP, Bermúdez-Cruz RM, Martínez-Peñañiel E, Guarneros G, Kameyama L. (2023) Early antitermination in the atypical coliphage mEp021 mediated by the Gp17 protein. *Archives of Virology* 168(3):92. doi:10.1007/s00705-023-05721-w

Bravo CE, **Valencia TG**, Tapia JR, Roldan EC, Calderon AA. (2014) Decolorization of azo dye by enteropathogenic Escherichia coli strains isolated from wastewater. *Journal of Pure & Applied Microbiology*

8(2):37–42.

Book chapter: **Valencia Toxqui G** et al. (2013) Impact on health due to enteropathogenic E. coli strains in an effluent from Puebla city. Young Researchers Presentations, I.N.I.C.E., Spain. ISBN 978-84-937854-5-1.

Selected Presentations (15+ oral & poster, 2012–2025)

2025. Cracking the code of phage Mu lysis (poster), Student/Postdoc Research Conf., College Station, TX.	2024. Lysis of phage Mu: the release of its endolysin (poster), ASM Texas Branch, Cedar Hill, TX.
2025. A global regulator reshapes host transcription to modulate lysis timing (poster), Molecular Genetics of Bacteria & Phages, Madison, WI.	2023. Comparative genomics of the mEp021 group (poster), Molecular Genetics of Bacteria & Phages, Madison, WI.
2025. Gp24 as a regulator of lysis timing in Mu phage (oral), Microtalks, College Station, TX.	2023. Functional characterization of Gp17 (poster), ASM Microbe, Houston, TX.
2024. Characterization & genome engineering of phages infecting uropathogenic E. coli (poster), Viruses of Microbes, Cairns, Australia.	2019. Characterization of gp17 (oral), Int'l Congress of Basic & Applied Microbiology, Puebla, MX.

Awards • Teaching • Mentoring • Service

Awards	ArtSci Research Award, Travel Program for Postdoctoral Researchers (2024); Travel Awards, TAMU Office of Postdoctoral Affairs (2024, 2026).
Teaching	Teaching Assistant, BICH-464 Bacteriophage Genomics (Spring 2025, Spring 2026); BIOL-100 Horizons in Biology (Fall 2025).
Mentoring	6 undergraduates (Ramsey Lab); 3 undergraduates & 4 master's students (Kameyama Lab).
Service	Judge, SACNAS Symposium (2024, 2025) & Student Research Week (TAMU, 2024); Undergraduate Research Scholars thesis reviewer; peer reviewer for BMC, PeerJ, FEMS Microbiology Letters, G3, and Recursos Naturales y Sociedad. Outreach: Darwin Day (2024, 2025).
Training	NCFDD Faculty Success Program; Postdoctoral Mentoring Academy; Advanced Teaching Practices for Higher Education; 6th International Viromics Workshop (2023).
Memberships	American Society for Microbiology (ASM & ASM Texas Branch); International Society for Viruses of Microorganisms (ISVM); AAUW; SACNAS.